

decode labs-python-internship

Python Internship Projects — DecodeLabs Batch 2026

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Projects built during my Python Developer internship at DecodeLabs.

■ Project 1 — To-Do List

- Concepts Used: Lists, `append()`, `for` loops, `enumerate()`, JSON persistence
- Features:
 - Add tasks
 - View all tasks with status
 - Mark tasks as Done
 - Delete tasks
 - Data saved to disk (`tasks.json`)
- How to Run: `python todo_list.py`

■ Project 2 — Expense Tracker

- Concepts Used: Accumulators, `while` loop, `try/except ValueError`, float type casting, sentinel values, JSON persistence, category breakdown
- Features:
 - Add expenses with category & description
 - Quick-add session (continuous entry mode)
 - Real-time running total using accumulator pattern
 - Category-wise breakdown with visual bar
 - Defensive coding — rejects invalid input
 - Data saved to disk (`expenses.json`)
- How to Run: `python expense_tracker.py`

■ Project 3 — Random Password Generator

- Concepts Used: `import string`, `import secrets`, `".join()`, entropy math $E=L*\log_2(R)$, NIST guidelines, JSON persistence
- Features:
 - Cryptographically secure via `secrets` module (NOT random)
 - Guaranteed character types (uppercase, lowercase, digits, symbols)
 - Entropy analysis in bits with crack-time estimate
 - NIST compliant (8-64 character range)
 - $O(N)$ efficient `".join()` — not $O(N^2)$ string concatenation
 - Generation history saved to disk
- How to Run: `python password_generator.py`

Built with ■ by Khushal Parmar